

Smart Traffic Lights Using Local AI

Shachi Kembhavamath
Department of Computer Science and Engineering
College Name
India
Email: example@email.com

Abstract—Smart traffic-light systems are being studied as a way to reduce waiting time and respond more quickly to changing road conditions. Traditional traffic lights usually follow fixed timers, while modern systems can use cameras and artificial intelligence to estimate traffic conditions and adjust signal timing. This paper presents a simple local-AI architecture in which traffic information is processed closer to the road. The proposed approach also considers emergency-vehicle detection as a research gap for future development.

Index Terms—Smart Traffic Lights, Local AI, Artificial Intelligence, Machine Vision, Traffic Signal Control, Emergency Vehicle Detection

I. INTRODUCTION

Smart traffic-light systems are being studied as a way to reduce waiting time and respond more quickly to changing road conditions. Traditional traffic lights usually follow fixed timers. They are simple and inexpensive, but they cannot easily react when one road suddenly becomes crowded.

Recent systems use cameras and artificial intelligence (AI) to estimate how many vehicles are present and then adjust the lights. Research reviews show growing interest in AI-based traffic signal control and its possible benefits for changing traffic conditions [1].

Cloud-based AI systems send camera information to a central server for processing. This can provide strong computing power, but it depends on an internet connection and can introduce communication delay. Local smart-camera systems move some processing closer to the road. This can reduce the amount of information that must be sent over the internet and may allow faster decisions.

Machine-vision research has also explored using camera-based vehicle information to adapt traffic-light timing [2].

II. MATHEMATICAL EQUATION

A simple traffic-flow measurement can be represented using the vehicle density equation:

$$D = \frac{N}{L} \quad (1)$$

where D represents vehicle density, N represents the number of vehicles, and L represents the road length considered.

Equation (1) provides a simple way to represent the relationship between the number of vehicles and the road section being considered.

III. TRAFFIC DATA ARRAY

Traffic information can be represented using a matrix. For example:

$$A = \begin{bmatrix} V_1 & V_2 & V_3 \\ 10 & 15 & 20 \\ 5 & 8 & 12 \end{bmatrix}$$

Here, the matrix can represent traffic measurements collected from different road sections.

IV. SYSTEM COMPARISON

The systems can be compared using decision time, internet-data usage, and setup cost.

TABLE I
COMPARISON OF TRAFFIC-LIGHT SYSTEMS

System	Decision Time	Internet Data	Cost
Basic Timed Lights	30 s	Zero	\$200
Cloud AI Systems	2 s	High	\$2500
Local Smart Cameras	0.5 s	Low	\$1200

The values in Table I are the illustrative comparison provided in the supplied handout.

V. SYSTEM IMAGE

Figure 1 can be used to show an illustration of the proposed smart traffic-light system.

VI. SYSTEM ARCHITECTURE

The proposed architecture follows four steps. First, street cameras capture traffic conditions. Second, the video information is sent to smart-pole computers, which perform local processing and make a quick traffic decision. Third, selected information can be forwarded to the city cloud server for storage and wider analysis. Fourth, the decision is sent to the traffic lights.

The overall flow is:

Street Cameras → Smart Pole Computers → City Cloud Server → Traffic Lights

VII. FLOWCHART

Figure 2 shows a simple flowchart of the proposed process.

VIII. ALGORITHM

Algorithm 1 presents a simple traffic-control procedure.

image.png

Fig. 1. Smart Traffic-Light System

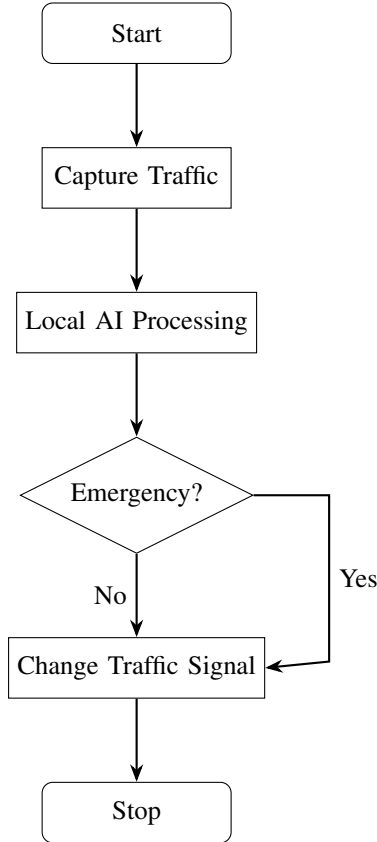


Fig. 2. Flowchart of the Proposed Traffic System

Algorithm 1 Local AI Traffic Control

```
1: Capture traffic-camera data
2: Detect vehicles from the camera input
3: Estimate traffic condition
4: Check for emergency vehicle
5: if emergency vehicle detected then
6:   Coordinate nearby traffic signals
7: else
8:   Adjust signal according to traffic condition
9: end if
10: Send decision to traffic-light controller
```

IX. RESEARCH GAP

Existing traffic systems can work well for regular daily traffic, especially when traffic patterns are predictable. However, an important emergency problem remains: many systems do not have a built-in way to automatically clear a path for a speeding ambulance.

This creates a need for a smart traffic system that can detect an emergency vehicle and quickly coordinate nearby signals to create a clear route.

X. CONCLUSION

Smart traffic-light systems can use camera-based information and local processing to respond to changing traffic conditions. Research on AI-based traffic signal control and machine vision provides relevant foundations for this area [1], [2].

The proposed architecture uses street cameras, smart-pole computers, a city cloud server, and traffic lights. Future work can investigate automatic emergency-vehicle detection and coordination of nearby traffic signals to create a clear route.

REFERENCES

- [1] F. Rasheed, K.-L. A. Yau, R. M. Noor, C. Wu, and Y.-C. Low, "Deep reinforcement learning for traffic signal control: A review," *IEEE Access*, vol. 8, pp. 208 016–208 044, 2020.
- [2] V. Gorodokin, S. Zhankaziev, E. Shepeleva, K. Magdin, and S. Evtyukov, "Optimization of adaptive traffic light control modes based on machine vision," *Transportation Research Procedia*, vol. 57, pp. 241–249, 2021.